

# RapidBoxForest: Link-Envelope-Guided Box Forests for Fast Multi-Query Manipulation

TIAN, Yu and Ren, Hongliang

**Abstract**—Repeated-query manipulation needs reusable free-space structure, yet constructing high-quality convex regions can be expensive in high-dimensional scenes and fragile under local edits. This paper presents RapidBoxForest (RBF), a link-envelope-guided box-forest planner for this low-build-cost regime. RBF repurposes the classical endpoint-bounding pipeline from swept-capsule collision detection as a configuration-space box-validation oracle: endpoint interval envelopes over a joint box are radius-lifted into link envelopes, which validate volumetric regions rather than individual path segments. Accepted boxes are grown into a scene-specific forest, consolidated into a typed corridor graph, and queried by membership lookup plus graph search. A Lifelong Envelope Cache Tree (LECT) separately stores kinematics-keyed endpoint and link-envelope evidence, while the scene-level RBF cache stores validated boxes and connectivity for one environment. The resulting planner trades regional expressivity for inexpensive box validation, simple maintenance, and explicit reuse boundaries. Its formal claim is soundness for paths contained in conservatively validated box unions; filters and post-processing are accepted through the configured query-validation path. Experiments on Shelf+IIWA tasks, saved random multi-robot scenes, mechanism studies, and local obstacle updates quantify the resulting build/query trade-offs and the benefit of reusable scene-level structure.

**Index Terms**—Motion planning, multi-query planning, link interval envelopes, box-region planning, configuration-space planning.

## I. Introduction

High-DOF manipulators often face many related queries in nearly fixed workcells: the robot kinematics persist, while start-goal pairs, local obstacles, or small scene edits change. Reuse is valuable only if the reusable object is cheap enough to build, update, and query.

Convex-region methods such as IRIS and Graphs of Convex Sets provide powerful region-level planning abstractions once certified regions are available [1], [2], [3], but their construction cost can dominate in high-dimensional or frequently edited scenes. Sampling-based methods such as PRM and RRT variants avoid heavy preprocessing and solve single queries effectively [4], [5], [6], [7], yet their reusable structures are usually zero-volume roadmaps or trees.

RBF targets the middle ground: it grows axis-aligned  $C$ -space boxes, validates them with a link-envelope scene oracle, connects accepted boxes into a graph, and answers later queries by searching the resulting corridor. The design trades region expressivity and path optimality for low build cost, simple adjacency, and explicit multi-query reuse. The resulting planning object is intended for regimes in which repeated-query latency and rebuild cost matter more than obtaining a tighter regional decomposition.

*a) Problem statement.*: Let  $\mathcal{Q} \subset \mathbb{R}^n$  be the joint-limit box of a fixed manipulator and let  $\mathcal{O} \subset \mathbb{R}^3$  be the current workspace obstacle set. Given a batch or stream of start-goal queries  $(q_s, q_g) \in \mathcal{Q}^2$ , the objective is to build a scene-specific planning object that supports repeated query retrieval while amortizing geometric work that depends only on robot kinematics and joint intervals. The object sought by RBF is a finite family of validated  $C$ -space boxes and an adjacency graph over those boxes. Its construction should have lower rebuild and local-update cost than a high-quality convex-region decomposition, while still exposing volumetric free-space structure rather than only path-local collision checks.

The key geometric ingredient is not a new capsule swept-volume identity, but a new use of the classical endpoint-bounding pipeline. For rounded link models, RBF separates the swept capsule into the sweep of a zero-radius center segment and a final radius lift. Endpoint interval envelopes bound the two endpoint trajectories over a joint box; the convex hull of these endpoint envelopes, enlarged by the link radius, encloses the swept rounded link. RBF uses this bound as a  $C$ -space box free-space validation oracle. The same endpoint evidence can be exported as axis-aligned bounding boxes (AABBs), SupportHull records, or staged AABB-SupportHull filters. Conservative endpoint sources yield a conditional conservative box certificate, whereas tighter practical sources act as planning filters under the common query-validation policy.

The planning layer builds directly on this primitive. A sample-guided grower selects query-relevant frontiers, materializes local candidate boxes around projected seeds, and extends a connected forest of validated regions. This is a workload-biased adaptive cell decomposition in manipulator  $C$ -space. RBF has two reuse levels. The scene-level RBF cache is the constructed box forest and corridor graph for a fixed obstacle set; it directly answers many

TABLE I  
Qualitative comparison of reusable manipulation-planning paradigms.

| Family           | Reuse object      | Strength                 | Limitation               | Relation to RBF                                                                                 |
|------------------|-------------------|--------------------------|--------------------------|-------------------------------------------------------------------------------------------------|
| PRM / Roadmap    |                   | Multi-query graph        | Edge checks; zero volume | RBF reuses validated boxes rather than only edges.                                              |
| LazyPRM          |                   |                          |                          |                                                                                                 |
| RRTConnect / RRT | Query tree        | Fast discovery           | Little persistent reuse  | RBF turns frontier growth into reusable box growth.                                             |
| BIT* / FMT*      | Sample graph      | Anytime improvement      | Query-centric checking   | RBF prioritizes reusable low-cost regions.                                                      |
| IRIS / GCS       | Convex regions    | Optimization-ready graph | Costly inflation         | RBF uses cheaper boxes instead of rich convex sets.                                             |
| RBF              | Box forest + LECT | Volume reuse             | Conservative boxes       | CCD-style link envelopes validate $C$ -space boxes; scene and kinematic caches remain separate. |

start-goal queries in that environment. The lifelong envelope cache, LECT, memoizes scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals; it accelerates future builds or local updates, while free-space labels remain scene-dependent.

The evaluation is organized around this design point. It examines repeated-query Shelf+IIWA planning, evidence replay, link-envelope representation trade-offs across AABB, SupportHull, and staged cascades, and local maintenance under obstacle edits. The resulting claims are framed around build cost, multi-query reuse, envelope-representation trade-offs, and update behavior rather than around a universal ranking against all single-query planners.

RBF is therefore a planning-systems contribution enabled by a geometric primitive. The endpoint-to-link envelope supplies the  $C$ -space box-validation oracle; the scene-level box-forest cache, lifelong LECT evidence cache, and local update rules show how that oracle can be assembled into a reusable multi-query planning pipeline.

**The paper contributes RBF, a link-envelope-guided box-region planner that lifts classical swept-capsule endpoint bounds from collision detection to reusable  $C$ -space free-space construction.** Its main algorithmic components are:

- 1)  *$C$ -space box validation from endpoint-to-link envelopes.* RBF repurposes CCD-style endpoint bounds as a joint-box free-space oracle, with conservative sources for certificates and tighter staged sources for query-validated filtering (section III).
- 2) *A reusable box-forest planner.* Seed descent, leaf-sweep coverage, restricted refinement, frontier expansion, and consolidation produce a scene-level typed corridor graph of validated boxes (section V).
- 3) *Local-level cache reuse.* Scene RBF caches store the current environment's box forest, while LECT stores lifelong kinematics-keyed endpoint and link-envelope evidence reusable across builds (sections IV and V).
- 4) *Local maintenance under scene edits.* Deletions revalidate formerly blocked domains for promotion, and insertions invalidate conflicting boxes before bounded local refill (section V-G).

The rest of the paper develops the envelope oracle (section III), LECT reuse layer (section IV), box forest

and update rules (section V), and evaluation (section VI).

## II. Related Work

### A. Convex Region Planning and Graphs of Convex Sets

Convex region planners approach motion planning by constructing a certified cover of free space and then solving graph search or convex optimization on top of that cover. IRIS introduced the now-standard idea of computing large collision-free convex regions by alternating separating hyperplanes and region inflation [1]. Later work extended the same line into configuration space, improved region quality, or improved graph compactness through visibility-graph and clique-cover constructions [8], [9], [10]. Marcucci *et al.* made the Graph-of-Convex-Sets formulation particularly influential by showing that motion planning can be cast as optimization over a graph of certified convex regions [2], [3].

RBF shares the region-level objective of this literature but uses a different primitive. Its regions are axis-aligned boxes in  $C$ -space rather than general polytopes, so the cover is less expressive than a high-quality convex decomposition. The payoff is a lower-cost validation loop, simple adjacency bookkeeping, and direct local maintenance. The relevant comparison is therefore not whether boxes are intrinsically better than polytopes; it is whether a lower-cost region primitive can support repeated-query planning before the cost of tighter convexification is justified.

### B. Swept Volume Approximation and Capsule Geometry

Computing the swept volume of a moving rigid body has been studied extensively in computational geometry and computer-aided manufacturing (CAM) communities [11], [12]. For a convex body  $P = \text{conv}(v_1, \dots, v_m)$ , bounding each generator trajectory by a conservative outer set  $\hat{\Gamma}_i(I)$  yields

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right),$$

a bound that is implicit in oriented bounding box (OBB) tree construction for swept mesh primitives [13].

For capsule links this bound reduces to bounding only the two segment endpoints. The Proximity Query Package (PQP) exploits exactly this Minkowski decomposition ( $C = S \oplus B_2(r)$ ), sweep of  $C$  reduces to sweep of  $S$  plus

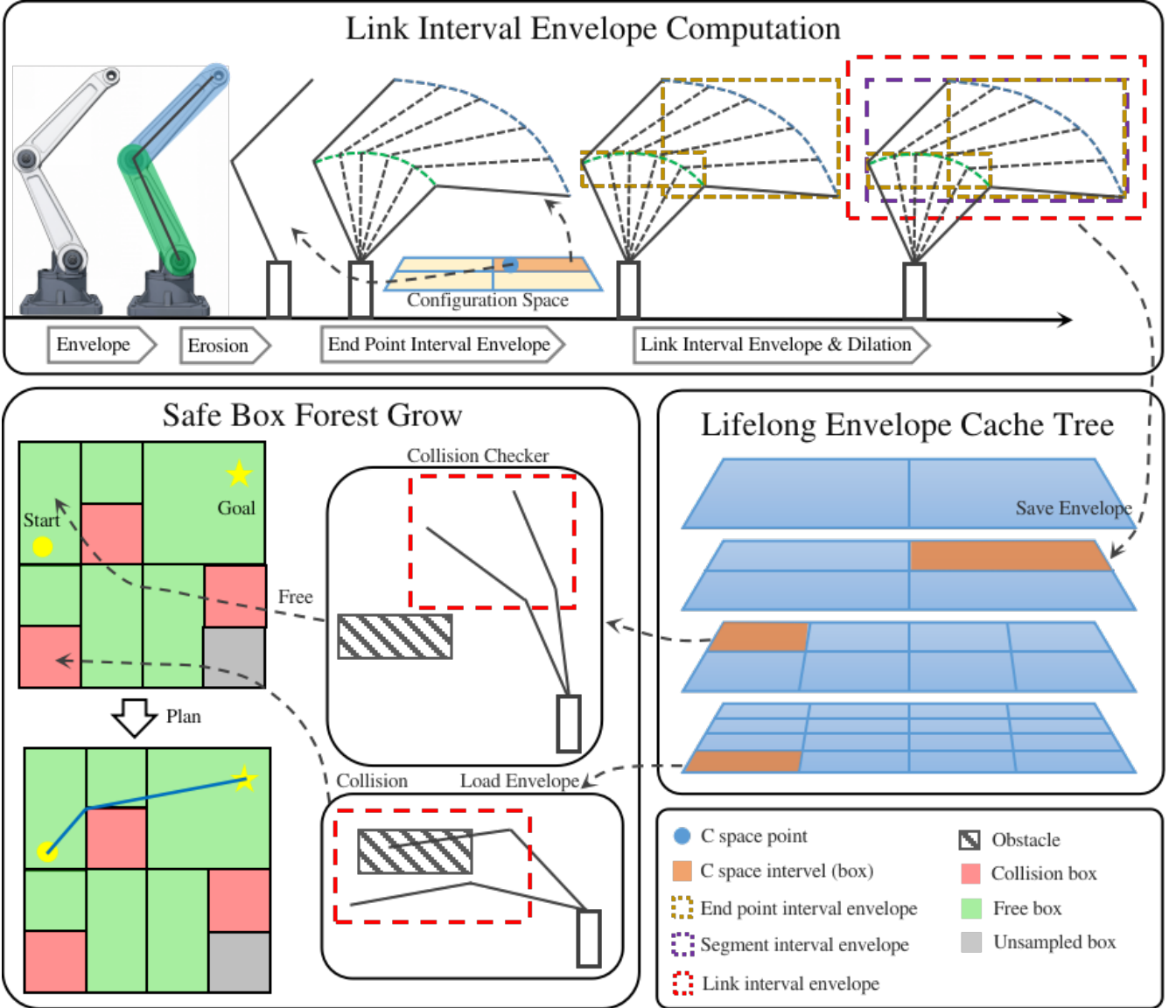


Fig. 1. RBF pipeline for link-envelope-guided multi-query planning. Top: endpoint interval envelopes induce link interval envelopes over a  $C$ -space box. Middle: LECT reuses kinematics-keyed endpoint and link-envelope records across repeated builds. Bottom: scene filtering and sample-guided growth expand a connected scene-level box forest for repeated corridor queries.

radius lift) [14]; Gilbert–Johnson–Keerthi (GJK) and its robust implementation by van den Bergen apply the same identity for distance queries [15], [16]; the Flexible Collision Library (FCL) generalises the swept-sphere interface to a full collision library [17].

### C. Interval Forward Kinematics and the CCD Endpoint-Bounding Pipeline

Interval arithmetic [18], [19] provides inclusion-monotone enclosures of real-valued functions over parameter intervals. Snyder applied interval arithmetic to parametric geometry to compute conservative bounding boxes for curves and surfaces without mesh sampling [20]. Merlet extended DH-chain interval propagation to certified workspace analysis for parallel and serial manipulators [21], [22].

These ideas meet directly in the continuous collision detection (CCD) literature for articulated bodies. Redon,

Kim, Lin, and Manocha [23] model each robot limb as a line-swept sphere (LSS, i.e. capsule) and propagate interval Denavit-Hartenberg (DH) transforms along the kinematic chain to obtain *interval vectors bounding the two LSS endpoint positions* over a motion time step. Algebraically, this is the radius-isolation identity  $C = S \oplus B_2(r)$  followed by endpoint bounding of the zero-radius segment  $S$  and a final radius lift. The resulting swept capsule is bounded by padding the endpoint bounding box with the link radius. Thus CCD already uses the two-step pipeline

$$\text{interval FK endpoint bounding} \longrightarrow \text{bbox}([\mathbf{p}_i] \cup [\mathbf{p}_{i+1}]) \oplus B_2(r) = \text{link swept-capsule bound}$$

which RBF reuses with a different parameter domain and planning role. The journal extension by Redon, Lin, Manocha, and Kim [24] applies the same framework to general robot arms and kinematic trees, making it a representative reference point for articulated-body CCD. Zhang

*et al.* [25] replace interval arithmetic with Taylor models in the same pipeline to reduce wrapping errors, analogous to how RBF’s HIFK source tightens the unsplit IFK bound by recursively subdividing joint boxes. Schwarzer *et al.* [26] address a multi-query motivation similar to RBF’s—reusing collision evidence across planning calls—but at the path-segment level rather than as volumetric  $C$ -space records. Corrales *et al.* [27] apply sphere-swept line (SSL) bounding volumes dynamically for human–robot safety monitoring using the same structural observation that bounding a capsule’s motion reduces to bounding its two endpoints.

#### D. Novelty Boundary: Prior CCD vs. RBF

The two-step endpoint-interval-to-swept-capsule pipeline, including radius isolation and final radius lift, is therefore already present in the CCD literature for time-parameterized motion. The contribution boundary in this paper is the use of that pipeline as a  $C$ -space free-region-construction primitive. The differences are: (a) the parameter domain is a  $C$ -space *joint-box interval* rather than a trajectory time step, so the goal is to validate a volumetric free-space region rather than to test a single path segment; (b) the endpoint source is a switchable certified interface—IFK\_AA and HIFK at different subdivision depths—rather than a fixed algorithm, allowing the tightness vs. cost trade-off to be controlled without changing the link-representation or planner layers; (c) compact per-representation link records (AABB and SupportHull) are organized for staged filtering during box growth, a design dimension absent from CCD libraries; (d) accepted boxes are assembled into a typed corridor graph whose overlap edges, tolerance-gap candidates, and segment witnesses have different certificate semantics; and (e) the endpoint and link-envelope records are organized in LECT, a persistent *kinematics-keyed* cache that reuses evidence across multi-query builds in related scenes, a reuse level not addressed by CCD libraries or path-segment caches. This distinction fixes the novelty boundary used throughout the paper. CCD uses the endpoint-bounding pipeline to enclose the swept volume of a body over a given time interval. RBF uses the same geometric pipeline over a joint box to validate candidate volumetric free-space regions, stores compatible evidence for replay, and organizes accepted regions into a queryable planning graph. Link envelopes therefore reduce repeated geometric work during free-region construction, while the query validation remains the acceptance rule for post-processed paths.

#### E. Multi-Query Reuse and Adaptive Configuration-Space Decomposition

Multi-query planners typically amortize effort by reusing a graph structure such as a roadmap. PRM is the classical example: build a connectivity graph once and reuse it for many start-goal pairs [6], [28]. LazyPRM, lazy edge checking, and certificate-based

planners postpone or cache edge validation so that online search does not repeatedly evaluate the same discrete local connections [29], [30]. Experience-based and trajectory-library systems reuse prior solutions or local motion snippets, often as initialization or bias for a new query rather than as a conservative decomposition of a region of configuration space [31].

RBF separates reuse into two levels. Within a fixed scene, the constructed box forest and corridor graph are the reusable planning object, so later queries reuse volumetric boxes rather than only paths, binary collision outcomes, or zero-volume edges. Across related builds or local scene edits, LECT can replay kinematics-keyed envelope evidence while the scene-dependent free-space decision is recomputed. PRM-like methods reuse connectivity and must revalidate affected edges after scene changes. Convex-region planners reuse higher-quality regions, but those regions are usually coupled to the obstacle layout in which they were constructed. RBF makes the boundary explicit: scene-level boxes are reused inside one environment, while kinematic envelope records can outlive that environment.

The method also inherits the central idea of adaptive cell decomposition and subdivision planning: represent free space by cells, split cells where the current evidence is insufficient, and search the adjacency graph induced by the accepted cells [32], [33]. The difference is the operating regime. Classical cell-decomposition examples are often low-dimensional workspace or explicit  $C$ -space constructions, whereas RBF applies the same adaptive-cell idea to high-dimensional manipulator  $C$ -space through a link-envelope oracle. Scene-specific obstacle filtering is layered on top of reusable kinematics-keyed envelope evidence, with LECT available only as a cache for expensive records and the scene-level forest serving as the direct multi-query cache. The trade-off is explicit: RBF accepts axis-aligned boxes and over-approximation in exchange for simple validation, adjacency maintenance, and repeated-query throughput.

#### F. Sampling-Based Planning and Path Refinement

Sampling-based planners remain standard baselines when the goal is to solve a single query quickly. RRT-Connect is particularly effective in manipulator planning because bidirectional growth often succeeds with very little global structure [34]. Asymptotically optimal and informed variants such as RRT\*, Batch Informed Trees (BIT\*), and Fast Marching Tree (FMT\*) improve path quality and search efficiency when more time is available [7], [35], [36], [37]. The Open Motion Planning Library (OMPL) has made these methods easy to deploy and benchmark in a common software stack [38], [39]. However, standard RRT variants produce single-use zero-volume evidence and typically restart from scratch on the next query.

By contrast, RBF uses sample-guided, frontier-biased expansion to materialize  $C$ -space boxes from the envelope

oracle, producing a volumetric corridor structure that can be queried repeatedly within the same environment. Trajectory optimizers address a different stage of the pipeline. CHOMP, STOMP, TrajOpt, and Gaussian-process motion planners refine an initial path by local descent or sequential convex optimization [40], [41], [42], [43]. RBF is complementary to these methods: it supplies a reusable region graph and leaves local smoothing, repair, and path checking to the downstream query pipeline.

Table I summarizes the resulting design trade-offs. The comparison is qualitative: it identifies what type of object each family reuses and where RBF fits in the chain from geometric evidence to  $C$ -space free-region planning, rather than ranking methods by the experimental results reported later.

### III. Link Interval Envelopes

This section defines the geometric primitive used by RBF: a reusable map from a joint box to compact link interval envelopes. The goal is not exact swept volume computation, but a low-cost outer representation for box growth. Strict endpoint sources give conservative certificates; tighter sampled or support-based sources act as filters before query validation.

Let  $\mathcal{Q}$  be the joint-limit box,  $I \subseteq \mathcal{Q}$  a joint interval, and  $\mathcal{O} \subset \mathbb{R}^3$  the obstacle set. Endpoint envelope  $E_k(I)$  outer-bounds endpoint  $k$ , and link envelope  $B_\ell(I)$  outer-bounds link  $\ell$  over  $I$ , yielding the primitive  $I \mapsto \{E_k(I)\} \mapsto B_\ell(I)$ . Conservative endpoint evidence gives a conservative link envelope; empirical evidence is used only as a filter.

**Assumption 1** (Collision model for certificates). *The formal certificates in this paper concern collision between the configured active rounded links and the current workspace obstacle set  $\mathcal{O}$ . Joint limits are enforced by  $I \subseteq \mathcal{Q}$ . Self-collision constraints, attached objects, or task-specific forbidden regions can be included by adding the corresponding link-obstacle or link-link checks to the same scene-validation policy; if they are not enabled, they are outside the certificate stated below.*

#### A. Geometric Foundation

Let  $T(q)x = R(q)x + t(q)$  denote the rigid transform induced by a configuration  $q \in I$ . For a reference body  $C_0 \subset \mathbb{R}^3$ , its pose at  $q$  is  $C(q) = T(q)C_0$ . We write  $S_I(C_0)$  for the exact workspace set swept as  $q$  ranges over the joint box  $I$  and  $\mathcal{C}_I(C_0)$  for its convex envelope. RBF usually works with the convex envelope because the downstream representations are AABBs or support-function hull records. The construction below deliberately reuses two classical facts—capsule Minkowski decomposition and convex-hull generator bounding—that already appear in swept-sphere collision libraries and articulated-body CCD work [23], [24]. The change is the domain and downstream object: the parameter set is a  $C$ -space joint box, and the output is a candidate box-region for a planning graph.

*Radius isolation.* For a capsule  $C = S \oplus B_2(r)$ , rigid motion commutes with the fixed Euclidean ball, giving exact two-stage sweep bounds:

$$S_I(C) = S_I(S) \oplus B_2(r), \quad \mathcal{C}_I(C) = \mathcal{C}_I(S) \oplus B_2(r).$$

RBF therefore follows the same centerline-first, radius-lift construction used by swept-sphere collision methods. This identity is exact for capsules; for sharp polytopes it requires the shape to be a rounded offset body.

*Convex-hull generator bound.* For a convex body  $P = \text{conv}(v_1, \dots, v_m)$ , bounding each vertex trajectory by an outer set  $\hat{\Gamma}_i(I)$  suffices to bound the swept polytope:

$$S_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right).$$

For a capsule link the seed is the center segment  $[a, b]$ , so only two endpoint trajectories need to be bounded.

Combining these two facts yields the practical recipe. For a capsule link  $[a, b] \oplus B_2(r)$ , RBF bounds only the two endpoint trajectories over the joint box, forms their convex hull, and applies the radius lift:

$$\hat{\mathcal{C}}_I([a, b] \oplus B_2(r)) = \text{conv}(\hat{\Gamma}_a(I) \cup \hat{\Gamma}_b(I)) \oplus B_2(r).$$

The numerical task reduces to computing reusable envelopes for two endpoint trajectories per link over  $I$ . Those records are then used by the planner to classify  $I$  as a candidate box-region, not merely to answer a path-segment CCD query.

**Proposition 1** (Endpoint envelopes induce link interval envelopes). *Consider a rounded link  $L = [a, b] \oplus B_2(r)$  and a joint interval  $I$ . If endpoint envelopes  $E_a(I)$  and  $E_b(I)$  satisfy  $\Gamma_a(I) \subseteq E_a(I)$  and  $\Gamma_b(I) \subseteq E_b(I)$ , then*

$$S_I(L) \subseteq \text{conv}(E_a(I) \cup E_b(I)) \oplus B_2(r).$$

*Consequently, any outer representation of the right-hand side is a valid link interval envelope for the whole joint box  $I$ .*

*Proof.* For each  $q \in I$ , the center segment of the moved link is the convex hull of the two moved endpoints, so  $T(q)[a, b] \subseteq \text{conv}(\Gamma_a(I) \cup \Gamma_b(I))$  over the entire interval. The endpoint inclusions place this set inside  $\text{conv}(E_a(I) \cup E_b(I))$ . Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment.  $\square$

**Corollary 1** (Conservative joint-box validation). *Let  $I \subseteq \mathcal{Q}$  be a joint box and let  $\{B_\ell(I)\}$  be conservative link interval envelopes for all active rounded links. If  $B_\ell(I) \cap \mathcal{O} = \emptyset$  for every link  $\ell$ , then every configuration  $q \in I$  is collision-free with respect to the collision model in assumption 1.*

*Proof.* By proposition 1, each conservative link envelope contains the swept workspace volume of its link over all configurations in  $I$ . Disjointness from  $\mathcal{O}$  for every active link therefore excludes workspace obstacle collision for the whole manipulator at every  $q \in I$  under assumption 1.  $\square$

#### B. Endpoint Interval Envelopes

Figure 1 illustrates the AABB extraction pipeline. Endpoint interval axis-aligned bounding boxes (iAABBs) are

computed by interval forward kinematics (IFK) or tighter certified AA-based variants, combined into a conservative zero-radius centerline envelope, and padded by the link radius before collision querying. The remaining numerical task is to outer-bound each endpoint trajectory over  $I$  by a reusable set  $E_k(I)$ . Endpoint envelopes are the common interface between kinematic propagation and the downstream link-envelope representations.

For endpoint  $k$ , let  $\Gamma_k(I)$  denote its exact workspace trajectory over interval  $I$ . Any conservative set that contains  $\Gamma_k(I)$  is an endpoint interval envelope. The basic endpoint record stores this envelope as an axis-aligned bounding box, because boxes are efficient to store, easy to combine, and directly reusable by all downstream link representations.

*a) Interval forward kinematics (IFK):* The IFK endpoint source used by the planner is backed by affine-arithmetic forward kinematics. It propagates workspace bounds along the DH chain and provides the lowest-cost certified member of the endpoint-source family. Its limitation is the usual wrapping effect: on wide boxes or long chains, one evaluation accumulates residual over-approximation across the full joint box.

*b) Hierarchical IFK (HIFK):* HIFK further reduces wrapping by recursively bisecting the joint box and evaluating AA-FK at the leaves before hulling the child endpoint boxes. HIFK is therefore a distinct endpoint source rather than a naming variant of unsplit IFK: a depth-1 HIFK evaluation already splits once and hulls two leaves, whereas IFK performs one unsplit AA-FK pass on the original box. The split depth is an endpoint-source parameter: larger depths reduce wrapping at the cost of additional FK evaluations.

*c) Incremental FK state reuse:* Tree descent also carries a transient FK state: interval-valued prefix transforms  $[\mathbf{T}^{0:0}], \dots, [\mathbf{T}^{0:n}]$  and the per-joint DH interval matrices used to build them. The full prefix chain is computed once at the root or after a restart; when only  $I_k$  changes, prefixes before  $k$  remain valid and only the suffix is recomputed. This short-lived state is not the persistent cache itself; it only keeps endpoint materialization efficient before LECT decides whether to retain the evidence.

Taken together, these sources define a clear certified spectrum: IFK\_AA is the lowest-cost source, while increasing HIFK depth offers progressively tighter outer bounds at increasing cost. Endpoint envelopes are the right intermediate abstraction for the planner because they isolate the kinematic bounding problem from the scene-query and graph-building problems. The downstream link-envelope layer remains fixed while the upstream source changes with the certification and runtime budget, and LECT can cache the resulting evidence without depending on the obstacle set.

### C. Link-Envelope Representations

Once endpoint envelopes are available, the remaining design question is how to represent the swept link volume

compactly for box growth, collision filtering, and optional reuse. For a capsule link, RBF does not reason about the thick link directly. It first bounds the zero-radius center segment between the two endpoint envelopes and then pads that centerline envelope by the link radius. A conservative link interval envelope  $B_\ell(I)$  is any set that contains the full swept capsule over interval  $I$ . In the box-based realization used below this is written as  $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$ , where  $B_\ell^\circ(I)$  bounds the centerline sweep and the padding operator  $\mathcal{R}$  becomes  $B_\infty$  for AABB envelopes.

The following representations share the same endpoint evidence but preserve different amounts of downstream geometric detail.

*a) AABB envelope:* AABB envelopes are the simplest option. The standard LinkIAABB construction takes the two endpoint envelopes, wraps them in one box, and pads that box by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus B_\infty(r_\ell).$$

This representation can be tightened by subdividing the reference center segment into  $S$  pieces, bounding each piece separately, and retaining the union of the padded sub-envelopes. Larger  $S$  preserves more directional variation along the link while keeping reconstruction efficient once endpoint envelopes are available. The main caveat is that the subdivision must remain explicit: if all sub-envelopes are collapsed back into one global AABB, much of the gain disappears.

*b) SupportHull envelope:* When the endpoint envelopes are convex, their convex hull preserves directional structure that a single AABB discards. SupportHull is the compact runtime realization of that idea: instead of storing an explicit dense hull, it keeps the two endpoint AABBs and the link radius as a support-function record. For a query direction  $d$ , the support point is the better of the proximal and distal AABB support points, and collision refinement against an obstacle AABB uses a fixed-size Gilbert–Johnson–Keerthi (GJK) narrow phase with the obstacle inflated by the link radius and safety margin. This retains much of the convex-hull directional fidelity while keeping the record compact.

Thus the envelope layer is modular with respect to the upstream kinematics solver: endpoint records determine the link envelope, and the chosen representation sets the planning-time cost/tightness trade-off.

## IV. Lifelong Envelope Cache Tree

The planner can materialize envelopes online; LECT is an optional replay layer for repeated builds or expensive envelope representations. It stores scene-independent endpoint/link-envelope evidence keyed by robot kinematics and joint intervals. Obstacle tests, box labels, graph connectivity, and query outcomes remain scene-local.

LECT is therefore separate from the scene-level RBF cache. The forest and corridor graph answer multiple queries in one fixed environment; LECT sits below that cache and reuses compatible kinematic evidence during builds, rebuilds, or local refills.

### A. Cache Key and Split Policy

LECT is a dynamically grown binary KD tree over the joint-limit box  $\mathcal{Q}$ . Each node represents one joint interval and stores interval bounds, split metadata, cached endpoint evidence from which link envelopes can be reconstructed, and residency state for the runtime. The key property is that the stored evidence is *kinematics-keyed* rather than *scene-keyed*: later scenes may replay a materialized envelope record, but they must rerun their own scene-local obstacle tests. Nodes use a heap-style breadth-first indexing convention (0 at the root and  $2p+1, 2p+2$  for children), which gives a finite indexed tree with configurable maximum depth  $D_{\max}$ .

a) *Evidence/label separation.*: An envelope record is replay-compatible only when the robot model, endpoint source, envelope representation, split descriptor, link radii, and active-link set match the current build. Replaying such a record imports endpoint and link-envelope evidence, not a free-space label. Every scene must still test the reconstructed link envelopes against its own obstacle set before a box can be accepted into the forest. This separation is what allows LECT to accelerate repeated geometry materialization without becoming a stale collision database.

The split policy is configurable and canonical for a given cache descriptor. The implemented schedules include round-robin, widest-root, and AAFK-volume-min. A build fixes one descriptor so that replay compatibility is well-defined; alternative descriptors are treated as separate cache settings. For a parent interval  $I$ , let  $I_d^-$  and  $I_d^+$  be the two children obtained by splitting joint dimension  $d$  at its midpoint, and let  $\mathcal{V}(I) := \sum_{\ell} \text{vol}(B_{\ell}(I))$  be the cumulative downstream envelope volume. The AAFK-volume-min schedule chooses depth-wise split dimensions that minimize a worst-child envelope burden,

$$d^* = \arg \min_d \max \left\{ \sum_{\ell} \text{vol}(B_{\ell}(I_d^-)), \sum_{\ell} \text{vol}(B_{\ell}(I_d^+)) \right\}.$$

The resulting descriptor is depth-synchronous: all nodes at the same depth use the same scheduled dimension when that dimension is still wide enough, otherwise the runtime falls back to a valid widest dimension. Candidate dimensions may also be filtered by FK reachability, since splitting a joint beyond the last active frame that contributes to any link endpoint cannot tighten the endpoint envelopes.

b) *Sparse interval staging.*: The compressed runtime path does not require every intermediate heap node on a deep split path to be materialized. A materialized planning cell can instead be represented by its root copy, per-dimension target-grid interval  $[\iota_d^-, \iota_d^+)$ , split-count vector, and exact joint interval. This is a Patricia-style overlay on the depth-synchronous split policy: branching and terminal cells are indexed, while unvisited ancestors remain virtual. The staging record also carries the interval fingerprint and split-policy hash used by LECT's exact interval replay path. Thus evidence replay remains keyed by the robot-compatible joint interval through

`endpoint_for_box_exact`; if no exact record is available, the same interval is live-materialized and then validated against the scene. Scene labels are still stored only in the forest or partition layer, not in LECT.

### B. Runtime Use and Validation Boundary

LECT is queried by FINDFREEBOX, not by the graph search itself. Given a joint interval, the cache either replays endpoint evidence or materializes it online, then derives the requested link representation. Directional records are used as refinement evidence rather than as standalone free-space certificates: the runtime first evaluates padded link AABBs, then uses SupportHull/GJK only for survivors of the AABB test. When a node is split, child envelopes may refine the parent witness, but any compact parent record remains representation-dependent and is rechecked against the current obstacle set before a box can enter the forest.

### C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint evidence is the common reusable record, but simple IFK+AABB evidence can be lower-cost to recompute online than to reload. SupportHull records are stored only when their tighter filters are expected to amortize storage and lookup cost. Reuse accounting should therefore attribute cache hits only to compatible kinematic evidence replay and not rely on additional symmetry-transfer assumptions.

The prewarm phase follows the same separation. It materializes FK/link-envelope evidence only on the target leaf layer and reconstructs upper layers by child-hull propagation during checkpoint. This avoids charging direct FK at multiple depths for the same cache and keeps parent witnesses tied to the leaf records that generated them. Exact leaf evidence and propagated child-hull evidence are both treated as envelope records by the replay path; provenance does not create a separate reuse or validation mode.

In the warm-build path, a cache hit reconstructs the requested link-envelope representation from stored endpoint records and revalidates it against the current obstacles. Warm builds still regrow scene-specific boxes; persisted box labels are not imported as certificates. Obstacle checking, adjacency, scene-level query reuse, and reactions to changing  $\mathcal{O}$  remain in the forest routines. Fixed-replay studies can disable writeback, while cache-growth studies use a separate writeback policy.

## V. RapidBoxForest Construction

This section describes the RBF planning layer. The algorithm grows a scene-specific forest of  $C$ -space boxes and exports it as a typed corridor graph for repeated queries. This forest-and-graph pair is the scene-level cache: while the obstacle set is unchanged, new queries use membership lookup and graph search instead of regrowing the forest. Samples choose frontiers, the envelope oracle validates boxes, and accepted boxes extend the connected region graph. The link interval envelopes of section III and

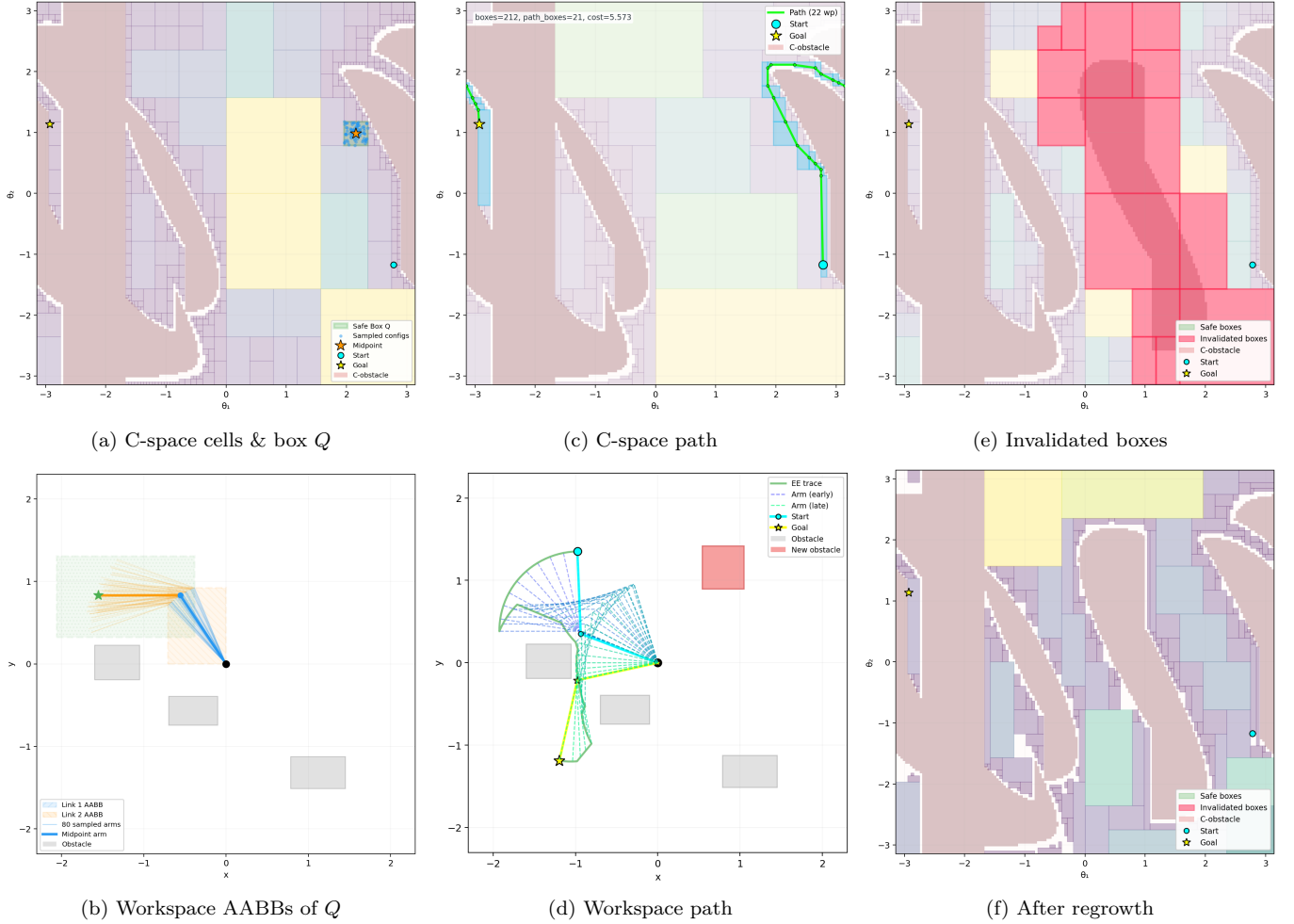


Fig. 2. Illustrative RBF build, query, and obstacle-update cycle on a planar 2-link manipulator. (a, b) Adaptive  $C$ -space cells and workspace AAB envelopes of the highlighted box  $Q$ . (c, d) Box-corridor path in  $C$ -space and its workspace trajectory. (e, f) After obstacle insertion, intersecting boxes are invalidated (red), and localized refill plus graph repair updates the explicit forest without requiring a full rebuild.

LECT of section IV provide the kinematic evidence used by this scene-level decomposition.

*a) Box, cell, and domain terminology.*: In the planner, a *box* is an axis-aligned joint interval  $B = \prod_d [l^{(d)}, u^{(d)}] \subseteq Q$ . The term *cell* is used when viewing the same object as an adaptive-decomposition element. A *collision domain* is a cell that has not been admitted to the forest under the scene-validation policy but is retained as a bounded refinement or update region. Thus boxes, cells, and domains share the same interval representation; the terms distinguish their role in the construction.

*b) Validation terminology.*: Throughout this section, a *conservatively validated box* is a joint box whose conservative link envelopes are disjoint from the current obstacle set. These boxes support the formal corridor statement below. A *performance-mode box* is accepted by a tighter nonconservative filter and is therefore treated as a planning candidate until the recovered path passes query validation. We use *validated box* for the box admitted by the selected scene-validation policy. This policy is the box-admission rule used during forest construction: it may be the conservative envelope-disjointness test, a performance-mode filter, or the same test augmented with

task-specific scene constraints. It is separate from the query-validation path, which checks recovered paths that do not inherit the conservative box-corridor certificate. We explicitly state when the stronger conservative certificate is required. Paths that leave the conservative box union through repair, smoothing, or external composition are charged where used and passed through the same query validation.

The build process has four stages. First, a seed-guided local oracle adaptively refines and classifies a  $C$ -space cell around a selected configuration. Second, the leaf-sweep mode constructs a shallow, reusable partition of the current scene and identifies conservative refinement domains. Third, repeated frontier expansion or restricted refinement grows validated cells into a connected forest, giving the method a region-valued, frontier-biased construction. Fourth, consolidation coarsens redundant boxes before the structure is exported as a query graph. The consolidation pass is part of the reusable build phase.

#### A. Construction Objective

Let  $Q$  denote the joint-limit box and let  $\mathcal{O}$  denote the current obstacle set. The construction objective is to build

a finite family

$$\mathcal{F} = \{B_1, \dots, B_N\}, \quad B_i \subseteq \mathcal{Q},$$

such that every  $B_i$  is admitted by the selected scene-validation policy, the union  $\bigcup_i B_i$  covers as much query-relevant free space as possible under the available box, time, and depth budgets, and the induced adjacency graph supports efficient downstream search. In conservative mode these admitted boxes are formal free-space certificates; in performance-oriented modes they are planning regions whose recovered paths require query validation. This object is scene-specific: changing  $\mathcal{O}$  changes which intervals are accepted into the forest, even though the underlying envelope evidence in LECT remains reusable.

The central structural rule is that the forest is not merely a free-space cover, but a *corridor-compatible* free-space cover. Boxes are inserted under a connected-box constraint.

Viewed as adaptive cell decomposition, this objective is intentionally workload-biased rather than globally exhaustive. The planner does not try to classify every cell in  $\mathcal{Q}$ . It refines and accepts cells near query-relevant frontiers, task seeds, and connector candidates, then keeps the resulting adjacency graph as the reusable planning object.

a) *Connected-box rule.*: For boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

RBF treats them as connected when, in every coordinate, the two intervals either overlap or are separated by at most a small tolerance  $\varepsilon_c$ :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

When  $\varepsilon_c = 0$ , this reduces to standard interval intersection or boundary touching in every dimension. A small positive tolerance absorbs the boundary offset used by grower seeds and floating-point roundoff.

This rule is deliberately weaker than requiring a full shared face. It treats overlap, touching, and tolerance-scale gaps uniformly as graph-connectivity evidence, while validity remains attached to the accepted box union itself. Only overlapping or touching box chains directly imply a contained corridor; tolerance-gap edges are query candidates that must be bridged by contained waypoints or by explicit segment validation. The output of the construction stage is therefore a forest of validated boxes whose graph connectivity is aligned with the corridor object queried later, without overstating a stricter face-sharing invariant than the grower actually maintains.

### B. Find Free Box

FINDFREEBOX descends LECT from the root toward the leaf containing seed  $q \in \mathcal{Q}$ , materializing endpoint and link envelopes lazily along the path, honoring reserved-node and skip-to-depth controls, and returning the first validated ancestor on the seed path. This is the largest canonical node on that path accepted by the scene-validation policy before further descent is needed.

**Algorithm 1** FindFreeBox: local box materialization around a seed.

---

**Input:** Seed configuration  $q$ , LECT  $T$ , obstacle set  $\mathcal{O}$ , depth cap  $D_{\max}$ , skip depth  $D_{\text{skip}}$   
**Output:** Validated interval  $I_v$  or failure

```

1:  $v \leftarrow \text{root}(T)$ ;  $d \leftarrow 0$ ; initialize current interval and FK state
2: while  $d \leq D_{\max}$  do
3:   Cache materialization
4:   materialize the endpoint and link envelopes of  $v$  if needed
5:   Reservation and scene validation
6:   if  $v$  is reserved and cannot be split further then
7:     return failure
8:   end if
9:   if  $d \geq D_{\text{skip}}$  and  $v$  passes the scene-validation policy then
10:    return interval  $I_v$ 
11:   end if
12:   Refinement
13:   if  $d = D_{\max}$  then
14:     return failure
15:   end if
16:   if  $v$  has no children then
17:     split  $v$  along the depth-synchronous split dimension of
       depth  $d$ 
18:   end if
19:    $k \leftarrow$  split dimension of  $v$ 
20:    $v \leftarrow$  the child of  $v$  whose interval contains  $q$ ; update the
     current interval
21:   update the FK state from dimension  $k$  onward;  $d \leftarrow d + 1$ 
22: end while
23: return failure

```

---

Tree topology and endpoint records are reused from the scene-independent cache, while scene-specific validation is performed on each grower visit. Algorithm 1 gives the pseudocode.

The termination behavior follows directly from the depth cap. Under  $D_{\max}$ , each loop iteration either returns immediately or descends one more level in the tree, so the procedure can make at most  $D_{\max} + 1$  decisions. In strict conservative mode, any returned interval has passed envelope disjointness against  $\mathcal{O}$ ; in performance-oriented modes, the returned interval is a validated planning region that remains subject to the common query-validation path.

### C. Adaptive Leaf Sweep and Restricted Refinement

The fast construction path uses FINDFREEBOX inside an adaptive terminal cell routine. Rather than relying only on stochastic frontier proposals to discover query-relevant coarse regions, RBF starts from a shallow depth-synchronous cell set and recursively classifies only cells that remain useful under the build budget. A terminal cell is one of five types: certified-free, certified-occupied, inconclusive-domain, deferred, or covered by an already accepted box. A certified-free cell is inserted into the initial forest and its descendants are pruned. An inconclusive cell is split only when its depth, blocker report, adjacency to accepted boxes, and seed evidence suggest that further refinement can improve the reusable cover. The fixed virtual leaf sweep is recovered as a special case that forces every inconclusive cell to split until the terminal depth.

During adaptive classification, RBF also records a blocker set

$$\beta(C) \subseteq \{1, \dots, |\mathcal{O}|\}$$

for each non-free or deferred collision domain. This set is conservative update metadata: it contains obstacle groups that were observed as blockers or were not proven free by the grouped shallow sweep. The blocker set is an over-approximate update explanation; certification of boxes entering the forest remains governed by the standard envelope validation rule.

An optional occupied predicate is used only when it can prove the whole cell is inside an obstacle. Otherwise the cell remains inconclusive or deferred.

**Lemma 1** (Signed-distance occupied pruning). *Let  $x_0$  be a material point on link  $\ell$ , let  $q_c$  be the center of cell  $I$ , and let  $\phi_O$  be the signed distance to obstacle  $O$  in workspace. If the workspace motion of this material point over  $I$  is bounded by  $\rho_{\ell, x_0}(I)$  and*

$$\phi_O(T_\ell(q_c)x_0) + \rho_{\ell, x_0}(I) + \epsilon_{\text{num}} < 0,$$

*then every configuration in  $I$  places that material point inside  $O$ , so  $I$  can be recorded as a certified occupied domain for pruning.*

When such a witness is unavailable, RBF does not infer occupancy; it keeps the cell as an inconclusive or deferred refinement domain.

After the shallow sweep, restricted refinement is performed only inside selected collision domains. Domains are ranked by their adjacency to already certified boxes, volume, and proximity to query-priority configurations. Candidate seeds are placed at interfaces between a collision domain and neighboring free boxes, with additional center and boundary seeds to avoid depending on a single contact point. For a domain  $C$ , every accepted refined box  $B$  must satisfy

$$B \subseteq C$$

and must be adjacent either to an existing forest box or to a box already accepted from the same domain. Thus the refinement stage can fill narrow passages revealed inside a shallow collision domain without spending its budget on unrelated regions of  $\mathcal{Q}$ . The connector stage may subsequently add collision-checked segment witnesses between remaining islands, but these witnesses are recorded separately from box-overlap edges.

If  $N_{\text{mat}}$  cells are materialized or replayed and  $N_{\text{term}}$  terminal cells are retained, the sweep work is  $O(N_{\text{mat}}(C_{\text{env}} + C_{\text{obs}}) + C_{\text{adj}})$  and the terminal storage is  $O(N_{\text{term}})$ , where  $C_{\text{env}}$  is the envelope materialization or replay cost,  $C_{\text{obs}}$  is scene validation cost, and  $C_{\text{adj}}$  is the indexed adjacency update cost. The important distinction from a fixed virtual layer is that  $N_{\text{mat}}$  is driven by accepted and frontier cells, not by the full cardinality of  $\mathcal{L}_{d_0:d_1}$ .

#### D. Forest Growth

Starting from an initial seed set, the forest is grown by repeatedly invoking the find-free-box procedure on carefully chosen candidate configurations. Each seed first produces a root box through the find-free-box procedure, and these root boxes initialize one or more connected

**Algorithm 2** AdaptiveLeafSweepRefine: terminal-cell coverage with restricted refinement.

---

**Input:** LECT  $T$ , obstacle set  $\mathcal{O}$ , sweep depths  $d_0, d_1$ , refinement budget  $N_r$   
**Output:** Validated forest  $\mathcal{F}$  and collision-domain cache  $\mathcal{C}$

- 1:  $\mathcal{F} \leftarrow \emptyset$ ;  $\mathcal{C} \leftarrow \emptyset$
- 2: initialize a priority frontier with start cells at depth  $d_0$
- 3: **while** frontier nonempty and build budget remains **do**
- 4:   pop the highest-priority cell  $C$
- 5:   **if**  $C$  is covered by an accepted box **then**
- 6:     mark  $C$  covered and continue
- 7:   **end if**
- 8:   materialize or replay LECT evidence for  $C$  and run scene validation
- 9:   **if**  $C$  is certified free **then**
- 10:     insert  $C$  into  $\mathcal{F}$  and prune descendants
- 11:   **else if** lemma 1 certifies  $C$  occupied **then**
- 12:     record  $C$  as a certified occupied terminal domain
- 13:   **else if**  $\text{depth}(C) = d_1$  or  $C$  is low-priority under no-good/connectivity rules **then**
- 14:     record  $C$  and its blocker set  $\beta(C)$  in  $\mathcal{C}$
- 15:   **else**
- 16:     split  $C$  by the configured split policy and push selected children
- 17:   **end if**
- 18: **end while**
- 19: build adjacency among boxes in  $\mathcal{F}$
- 20: **for all** collision domains  $C \in \mathcal{C}$  in priority order **do**
- 21:   generate interface, center, and boundary seeds inside  $C$
- 22:   **for all** seed  $q$  while budget remains **do**
- 23:      $B \leftarrow \text{FindFreeBoxInDomain}(q, T, \mathcal{O}, C)$
- 24:     **if**  $B \neq \emptyset$ ,  $B \subseteq C$ , and  $B$  is adjacent to  $\mathcal{F}$  or to a prior box from  $C$  **then**
- 25:       insert  $B$  into  $\mathcal{F}$  and update local adjacency
- 26:     **end if**
- 27:   **end for**
- 28: **end for**
- 29: **return**  $\mathcal{F}, \mathcal{C}$

---

components of the forest. The main grower is an evidence-guided, frontier-biased expansion process rather than a generic random cover. Each iteration chooses a frontier component, proposes a nearby interval worth testing, and accepts the result only when it extends the forest by an admissible connected corridor segment. This keeps the build process focused on boxes that can improve query connectivity rather than on uniformly covering all of  $\mathcal{Q}$ .

Each iteration makes two explicit heuristic decisions: it samples a target configuration  $q_t$ , then chooses a boundary seed on an existing box face that is closest to that target. The target sampler is a configurable categorical mixture over intertree roots, unexplored-volume targets, query roots, fixed anchors, and uniform roots. Build connectivity is handled primarily by overlap-compatible box insertion; segment bridges remain a secondary mechanism recorded separately from box-overlap edges. The mixture weights, depth caps, cache depth, thread count, and anchor set are planner settings rather than envelope-theory parameters.

Given a target  $q_t$  and an existing box  $B = \prod_j [\ell_j, u_j]$ , the grower enumerates feasible faces  $f = (B, d, s)$ , where  $d$  is a joint dimension and  $s \in \{-1, +1\}$  is the lower or upper side. Let  $\tau$  be the adjacency tolerance and  $\epsilon_b = \max(\epsilon_{\text{guard}}, \tau/4)$ . A face is feasible only if stepping outside it by  $\epsilon_b$  remains inside the global root interval. The seed

**Algorithm 3** GrowForest: budgeted expansion of a connected box-region forest.

---

**Input:** LECT  $T$ , obstacle set  $\mathcal{O}$ , seed set  $\mathcal{S}$   
 1: **Budgets/settings:**  $N_{\max}, t_{\max}, K_{\text{face}}, K_{\text{fail}}, H_{\text{cool}}$   
**Output:** Validated box-region forest  $\mathcal{F}$   
 2:  $\mathcal{F} \leftarrow \emptyset$ ; initialize connected components from  $\mathcal{S}$   
 3: **for all**  $q \in \mathcal{S}$  **do**  
 4:    $B \leftarrow \text{FindFreeBox}(q, T, \mathcal{O})$   
 5:   **if**  $B \neq \emptyset$  **then**  
 6:     insert  $B$  into  $\mathcal{F}$  as a root box  
 7:   **end if**  
 8: **end for**  
 9: **while**  $|\mathcal{F}| < N_{\max}$  and time budget not exhausted **do**  
 10:   draw target  $q_t$  from the configured connector/query/unexplored/uniform mixture  
 11:   enumerate feasible faces  $f = (B, d, s)$  of frontier boxes and compute  $q_f$  and  $\sigma(f; q_t)$   
 12:   rank the  $K_{\text{face}}$  lowest-score face seeds  
 13:   choose the first ranked seed that is not deferred and not in the frontier-seed cache  
 14:   set  $(q_{\text{seed}}, B_{\text{near}}) \leftarrow (q_f, B)$  for that face  
 15:    $B_{\text{new}} \leftarrow \text{FindFreeBox}(q_{\text{seed}}, T, \mathcal{O})$   
 16:   **if**  $B_{\text{new}} \neq \emptyset$  and  $\text{BoxesConnected}(B_{\text{new}}, B_{\text{near}}, \tau)$  **then**  
 17:     insert  $B_{\text{new}}$  into  $\mathcal{F}$ ;  $\text{UpdateComponents}(\mathcal{F})$   
 18:   **else**  
 19:     record the failed LECT leaf; defer it after  $K_{\text{fail}}$  failures for horizon  $H_{\text{cool}}$   
 20:   **end if**  
 21: **end while**  
 22: **return**  $\mathcal{F}$

---

associated with the face is

$$q_f[j] = \begin{cases} u_d + \epsilon_b, & j = d, s = +1, \\ \ell_d - \epsilon_b, & j = d, s = -1, \\ \text{clip}(q_t[j], \ell_j + \epsilon_b, u_j - \epsilon_b), & j \neq d, \end{cases} \quad (1)$$

with clipping to  $[\ell_j, u_j]$  when a box is thinner than  $2\epsilon_b$ . Faces are ranked by the squared target distance

$$\sigma(f; q_t) = \|q_t - q_f\|_2^2, \quad (2)$$

and the grower retains a bounded candidate set of size  $K_{\text{face}}$  across the scanned frontier boxes. It then tries them in increasing score order, skipping a candidate if the corresponding LECT leaf is temporarily deferred or if a component-connection cache has already recorded the same boundary seed. The first remaining candidate becomes the seed for FINDFREEBOX.

The candidate box is accepted only if FINDFREEBOX validates it and the new box is connected to its parent under the same adjacency predicate used by the query graph: all dimensions overlap up to tolerance  $\tau$ , and at least one dimension touches within tolerance or all dimensions overlap. Failures are also handled explicitly. When the LECT leaf containing a failed seed reaches the configured hard-frontier failure threshold  $K_{\text{fail}}$ , the leaf is skipped for a deferral horizon  $H_{\text{cool}}$  of subsequently accepted boxes, unless a later depth stage raises the active search depth and enables a retry. This deferral rule and the frontier-seed cache do not alter the validation boundary; they only reduce repeated calls on already failed or already covered frontier regions. The resulting method is summarized in algorithm 3.

The same construction also supports coordinated multi-goal growth. Multiple seed configurations can initialize roots in one authoritative forest, allowing the scene-level

**Algorithm 4** ConsolidateForest: graph-preserving merging and pruning.

---

**Input:** Validated forest  $\mathcal{F}$ , LECT  $T$ , obstacle set  $\mathcal{O}$   
**Output:** Consolidated validated forest  $\mathcal{F}'$   
 1:  $\mathcal{F}' \leftarrow \mathcal{F}$   
 2: *Stage 1: validated connected/contact merging*  
 3: merge exact connected pairs whose union is itself a single box  
 4: **for all** connected candidate pairs  $(B_i, B_j)$  in  $\mathcal{F}'$  **do**  
 5:   build a candidate parent region  $H$  and score its volume expansion  
 6:   **if** the chosen envelope representation of  $H$  passes the scene-validation policy **then**  
 7:     replace  $B_i, B_j$  by  $H$   
 8:   **end if**  
 9: **end for**  
 10: *Stage 2: containment pruning*  
 11: remove only contained boxes whose typed adjacency and segment witnesses transfer to selected covering regions  
 12: **return**  $\mathcal{F}'$

---

RBF cache to serve a batch of related queries while sharing LECT evidence across build workers and cache-warm reruns. Staged wavefront growth and task-batched growth are scheduling choices; they do not change the mathematical object produced by the planner, namely the query-ready box-region graph.

### E. Forest Consolidation

Raw growth tends to produce a forest that is finer than necessary. Consolidation reduces this representation without changing the validation rule or the typed-edge semantics used by corridor search. The planner uses two conservative operations. First, connected pairs whose union is again a box may be merged after validation. Second, more general connected candidate pairs are ranked by a volume-expansion score and accepted only if the candidate hull passes the same scene oracle. A final containment-pruning pass removes boxes that are fully covered by accepted validated regions when their graph incidence can be transferred. The consolidation logic is summarized in Algorithm 4.

Validity follows directly: every accepted merge has itself passed the scene oracle. Containment pruning removes only boxes fully covered by a selected region that inherits their typed adjacency and segment witnesses, preserving the connectivity semantics used by query search.

### F. Corridor Query

After growth and consolidation, the forest is exported as a typed graph  $G = (V, E_{\cap} \cup E_{\varepsilon} \cup E_s \cup E_{\pi})$  with one vertex per validated box. Edges in  $E_{\cap}$  connect boxes that overlap or touch in all dimensions; these edges directly preserve a contained box corridor. Edges in  $E_{\varepsilon}$  come only from tolerance-scale connected-box gaps and are treated as query candidates rather than as corridor certificates. Edges in  $E_s$  are explicit collision-checked segment witnesses. Edges in  $E_{\pi}$  are conservative portal edges: each stores a compressed edge between two global boxes together with a finite hidden chain of conservatively validated internal boxes. The chain is not materialized as global graph vertices, but it is expanded when the edge is

used by query path extraction. Because typed connectivity is maintained during growth and consolidation, online querying reduces to membership lookup and typed graph search rather than post-hoc graph reconstruction.

If disconnected adjacency islands remain after growth, an optional connector stage attempts targeted bidirectional sampling bridges between nearest frontier boxes of different components. A successful bridge is stored as a segment edge rather than as a claim that the two boxes overlap. This distinction is important for both the RBF query layer and the GCS negative controls: a segment edge is a collision-checked path witness between two boxes, whereas a GCS edge over convex regions requires feasible overlap of the regions themselves.

This pair  $(\mathcal{F}, G)$  is the planning object produced by the build stage. It is also the scene-level RBF cache for subsequent queries in the same environment. Given start and goal configurations  $q_s, q_g$ , the planner associates them with start-side and goal-side boxes in  $\mathcal{F}$  and then searches  $G$  for a connecting box sequence. The online query problem is reduced to box-corridor retrieval inside an already constructed box-region representation.

A shortest path on  $G$  yields a candidate box sequence between the two query endpoints. When the recovered geometric path stays inside the union of conservatively validated boxes, it is certified collision-free by theorem 1. Paths that use  $E_\varepsilon$  or  $E_s$  edges, performance-mode boxes, or post-processing steps that leave the conservative box union are treated through the configured query-validation path, and the query interface records these cases through the **QueryResult** counters. The box-corridor statement below applies to paths contained in that union. Portal edges in  $E_\pi$  are nonsegment edges: they inherit the conservative box-corridor certificate only through their stored internal box chain.

For an  $E_\cap$ -only graph path, this containment condition has a direct constructive interpretation. Each vertex box is convex, and each  $E_\cap$  edge means that consecutive boxes overlap or touch. Therefore a continuous piecewise-linear curve can be chosen by selecting one point in each nonempty pairwise intersection and connecting these points inside the corresponding boxes. If  $q_s$  and  $q_g$  lie in the first and last boxes, respectively, the same convexity argument connects them to the first and last intersection points. The resulting curve is contained in the union of the boxes on the graph path. This is a containment certificate, not a positive-clearance certificate: touching boxes may meet at a lower-dimensional set, and clearance depends on the underlying envelope-obstacle separation or on the configured query-validation test. This construction does not apply to  $E_\varepsilon$  or  $E_s$  edges, whose semantics are candidate gap closure and explicit segment witnessing, respectively.

**Corollary 2** (Conservative portal expansion). *Let  $(B_u, B_v) \in E_\pi$  store an internal chain  $B_u, B_1, \dots, B_m, B_v$  such that every internal box is conservatively validated and every consecutive pair overlaps or touches. Replacing the*

*portal edge by this chain recovers an  $E_\cap$ -only box-corridor segment. Therefore any query path using such portal edges is covered by theorem 1 after all portals are expanded.*

**Theorem 1** (Conditional conservative box-corridor path). *Let  $\mathcal{F}$  be a forest whose boxes have each passed conservative link-envelope validation against obstacle set  $\mathcal{O}$ . If a continuous path  $\gamma : [0, 1] \rightarrow \mathcal{Q}$  satisfies  $\gamma([0, 1]) \subseteq \bigcup_{B \in \mathcal{F}} B$ , then every configuration on  $\gamma$  is collision-free with respect to the collision model in assumption 1.*

*Proof.* For any  $s \in [0, 1]$ , the containment assumption gives a conservatively validated box  $B \in \mathcal{F}$  with  $\gamma(s) \in B$ . By construction, validation of  $B$  means every conservative link envelope for  $B$  is disjoint from  $\mathcal{O}$ . Because each such envelope contains the link geometry for every configuration inside  $B$  (corollary 1), the robot geometry at  $\gamma(s)$  is also disjoint from  $\mathcal{O}$ . Since  $s$  was arbitrary, the whole path is collision-free.  $\square$

a) *Reporting protocol.*: Theorem 1 is a conditional statement: it covers path portions contained in a conservatively validated box union. Post-processing inherits this certificate only when it preserves that containment condition. The result is therefore a *soundness* statement for the returned box-contained corridor, not an unconditional claim that a finite-budget forest covers every feasible passage in  $\mathcal{Q}$ . Coverage is controlled by the selected anchor set, subdivision depth, refinement budget, and update policy. In the conservative mode, any path retained inside the accepted conservative box union inherits the certificate above; paths that require uncovered regions must wait for additional refinement, a larger forest budget, or an explicit query-validation path.

### G. Dynamic Obstacle Updates

The explicit forest representation also supports local typed-graph maintenance under obstacle edits. The update procedure assumes fixed robot kinematics and a fixed LECT topology/evidence cache; only the obstacle set and scene-dependent forest labels and typed edges change. The key distinction is asymmetric: removing obstacles can expose previously blocked cached domains, whereas inserting obstacles may invalidate boxes and segment witnesses that were previously accepted.

The update invariant is local but explicit: every box retained or inserted after an edit must pass the edited-scene validation policy, every  $E_\cap$  edge must still correspond to overlap or contact of its endpoint boxes, every  $E_\varepsilon$  edge remains only a tolerance-gap query candidate, every  $E_s$  edge must keep a valid segment witness in the edited scene, and every  $E_\pi$  edge must retain a conservatively validated internal box chain or be removed. The update procedure is therefore a maintenance rule for the represented forest and its typed graph, not a claim of complete rediscovery of all newly available free-space regions after an edit.

For obstacle deletion, the blocker sets  $\beta(C)$  recorded during the leaf-sweep stage provide a conservative candidate-selection rule for the information that was

actually cached. When obstacle indices  $R$  are removed, each cached collision domain  $C$  updates its blocker set to  $\beta(C) \setminus R$ , with the remaining obstacle indices remapped to the edited scene. If this set becomes empty, the domain becomes eligible for promotion. Promotion either reuses sufficient cached disjointness evidence for the edited obstacle set or reruns the standard scene check on the reconstructed envelope record. It is also filtered by containment and adjacency checks: contained candidates are discarded, disconnected candidates remain cached, and accepted candidates are inserted into the forest and connected incrementally. Thus deletion updates use the recorded blocker cache to focus validation effort and are limited to domains represented in that cache. Discovering additional newly free regions is left to a subsequent build or refinement stage.

Obstacle insertion uses the complementary path. Each added obstacle is first tested against the current live boxes, optionally restricted by the spatial dirty region. Boxes that collide with the added-obstacle checker are removed from the forest and released from the oracle reservation map; raw boxes are checked and removed by the same edited-scene logic. Segment edges whose endpoints disappear or whose waypoint witness fails the edited-scene validation are removed as well. Each invalidated box then becomes a bounded refill domain. The insertion routine performs a local leaf sweep inside this domain up to the configured insertion depth, optionally capped relative to the invalidated box depth. Leaves that pass the scene-validation policy are reinserted; leaves that remain blocked or hit the depth cap are cached with a blocker set containing the new obstacle index. The insertion update is therefore a local invalidation-and-refill operation over the typed graph: invalidated boxes are removed, local leaf-sweep refill proposes replacement boxes, affected segment witnesses are revalidated, and typed connectivity is repaired incrementally. If the edited graph still cannot answer the query, the query pipeline may use its configured repair or segment-edge mechanisms.

If no contained conservative corridor is recovered, the box-corridor certificate does not apply; repaired, post-processed, or segment-recovered paths are accepted only through query validation.

## VI. Experiments

The experiments evaluate RBF as a reusable manipulation-planning system. They are organized as a dependency chain: the Shelf+IIWA ablation selects the RBF planning profile, the cross-algorithm and random-scene studies reuse that profile unless they explicitly sweep a budget parameter, and the dynamic-update study applies the same profile to edited scenes. All random scenes and query sets are saved before planner execution, so each method is evaluated on the same obstacle and query records. Tables and figures are generated from JSON/CSV artifacts with recorded provenance.

---

### Algorithm 5 DynamicUpdate: obstacle deletion promotion and insertion refill.

---

**Input:** Forest  $\mathcal{F}$ , typed graph  $G$ , collision-domain cache  $\mathcal{C}$ , edit  $(\Delta^-, \Delta^+)$   
**Output:** Updated forest, typed graph, and collision-domain cache

```

1: if  $\Delta^- \neq \emptyset$  then
2:   for all  $C \in \mathcal{C}$  do
3:     remove deleted obstacle indices from  $\beta(C)$  and remap surviving indices
4:     if  $\beta(C) = \emptyset$  then
5:       promote  $C$  to  $\mathcal{F}$  only if containment, adjacency, and edited-scene validation pass
6:     end if
7:   end for
8:   incrementally connect promoted boxes to overlapping or touching forest boxes
9: end if
10: if  $\Delta^+ \neq \emptyset$  then
11:    $\mathcal{I} \leftarrow \{B \in \mathcal{F} : B \text{ conflicts with some obstacle in } \Delta^+\}$ 
12:   remove  $\mathcal{I}$  and typed graph/segment witnesses that fail edited-scene validation
13:   for all  $B \in \mathcal{I}$  do
14:     set the local sweep depth, optionally capped relative to  $\text{depth}(B)$ 
15:     sweep leaves inside  $B$  to that local depth
16:     insert swept leaves that pass edited-scene validation and are adjacent to the surviving forest
17:     cache still-blocked leaves with blockers in  $\Delta^+$ 
18:   end for
19:   incrementally connect newly inserted boxes and revalidate surviving segment edges
20: end if
21: if the requested query is still disconnected then
22:   optionally run endpoint recovery and collision-checked segment recovery
23: end if
24: return updated  $(\mathcal{F}, G, \mathcal{C})$ 

```

---

a) *Common success and timing semantics.*: Planner success means that the returned path passes the fixed-resolution query-validation check. Validation time is reported separately and is not charged to planning time. Success rate is recorded over all scheduled runs; the main Tables II–IV omit the success-rate column because all selected entries are full-success points. Unless a caption states otherwise, reported time columns are the 25–75 percentile interval only, without printing a separate median value. Main path columns report 25–75 percentile intervals of success-only path-length ratios normalized by the best median path for the same saved query or scenario, avoiding a direct mix of geometrically different queries. Query-time shortcut boxes and final collision shortcut boxes are disabled for the RBF rows. Main online tables define Online/q as solver time only: endpoint anchoring, online repair, graph search, or an OMPL solve call. Final simplification is fixed globally to a 10 ms OMPL budget and is not included in Online/q or in amortized online-efficiency numbers. The independent strict audit remains reported separately. Appendix trade-off curves additionally sweep 0, 5, 10, and 50 ms simplification budgets, with 50 ms interpreted as a full-polish upper-budget setting rather than the primary online-efficiency number.

b) *RBF planning profile.*: Unless otherwise stated, RBF uses the profile selected in the Shelf+IIWA ablation: a query-agnostic offline build followed by reusable online queries. The offline phase constructs an adaptive grid-partition representation over the native joint-limit

space. Canonical symmetry is used only inside LECT; the planner inputs, outputs, query endpoints, and final audits remain in native joint coordinates. The selected shelf profile uses a parallel virtual leaf sweep from depth 8 to 13, d23 external endpoint evidence, budgeted grid merging, and partition-native adjacency/index construction. No start/goal, query waypoint, or query label is passed to the offline build, and the selected shelf profile does not add offline shortcut or segment edges. The online solve phase then probes the partition, applies query bridge/local repair to the five query pairs, and runs partition-native graph search; the fixed 10 ms OMPL simplification attempt is run afterward under the common final-processing protocol. The registered shelf profile fixes the b100 offline/bridge budget and keeps FINDFREEBOX binary descent with skip-to-depth 32, query-corridor paving depth 52, 12 forced bridge attempts with 1600 bridge-RRT iterations per online repair task, direct-corridor sample step 0.08 rad, and an adaptive repair cap of 24 calls. Support-hull collision uses an AABB broadphase followed by support-hull GJK narrow-phase validation. Query bridge repair is applied using the same global parameters for all five shelf query pairs. Residual segment witnesses, when inserted online after maximum-depth box repair fails, are explicitly accounted. Segment validation uses the same 0.01 rad maximum joint-space sample spacing as the final strict audit, so a segment witness cannot be accepted under a looser build-time collision check. Shelf+IIWA baseline rows additionally use the **Exp4-d23-b100** warm-cache setting, where the d23 external LECT evidence cache is read-only. Random-scene rows use the same algorithm defaults, robot-local LECTDB caches, and query-independent random offline anchors rather than Shelf-specific anchors.

#### A. Shelf+IIWA Leaf-Sweep-RRT Trade-off

Figure 3 and table II show the Shelf+IIWA b100 registered profile over seeds 0–7 and the five canonical query pairs. The table reports offline build time, batch-amortized online time, and build amortized over the five shelf queries at this fixed profile. CritSample rows are included only as endpoint-source diagnostics: they use the same planner and strict audit, but the envelope approximation is not a conservative-complete support-hull certificate.

#### B. Shelf Cross-Algorithm Context

The Shelf+IIWA cross-algorithm experiment uses the same five canonical queries and 0.01 rad final-audit setting for RBF, IRIS-NP+GCS, PRM, RRTConnect, and BIT\*. RBF rows use the Exp. 4 b100 reusable-planner profile in fig. 3; table III reports the same registered profile. For the per-query entries in table III, the RBF online latency is measured by a separate artifact that reruns each shelf query independently from the same query-agnostic profile; it is not obtained by dividing the five-query batch run or by summing overlapping bridge-task diagnostics. The Shelf+IIWA ablation in table II separately reports reusable batched amortization. PRM is reported as a

reusable build/query planner whose roadmap is shared across the five queries. RRTConnect and BIT\* are one-shot online baselines with zero build time. PRM, RRTConnect, and BIT\* are full-joint-space reruns under the same native-query semantics and final audit. IRIS-NP+GCS is restored as the legacy Shelf+IIWA contextual result. BIT\* uses one timed OMPL trace per query with `samples_per_batch=100`, `rewire_factor=5.0`, `stop_on_solution_improvement=false`, a 0.5 s per-query cap, checkpoints at 0.05, 0.1, 0.2, 0.3, and 0.5 s, and the same 0.01 s final OMPL simplification budget as the other OMPL reruns. The BIT\* table entry is selected from this checkpoint curve using the observed checkpoint return time, not the timeout cap; all reported paths still pass the same 0.01 rad strict final audit. The comparison is therefore an online-latency and amortization context for the reusable box-region graph, not a claim that RBF dominates one-shot planners in path length or single-query total time.

#### C. Random Multi-Robot Scenes

The random-scene experiment uses a saved q10x10 prefix catalog generated before planner execution and then reused by every evaluated planner. For each robot and scene seed, the retained medium and hard scenes share the same ten query pairs, and hard strictly extends medium with additional workspace obstacles. The original easy prefixes are kept in the saved catalog as diagnostics but are not reported here because their reference paths are dominated by direct-line solutions. The generator samples actual start and goal configurations in native joint space, stores canonical information only as LECT metadata, rejects direct-line trivial cases, and uses strict collision-checked reference-planner probes to validate the saved query set. Scenes are stratified by robot (IIWA, UR5, Panda), difficulty (medium, hard), and ten saved scene seeds, giving 600 saved query evaluations per planner method. Random-scene RBF rows build one query-agnostic forest per obstacle scene and then reuse it for the ten online queries. They do not reuse Shelf+IIWA anchors or the Shelf d23 cache; UR5 and Panda use robot-local d20 LECTDB caches, while IIWA uses a d23 LECTDB root. The random-scene scan follows the paper Exp.4/Exp.5 defaults (leaf max depth 13, deep FFB depth 62, connector/query-bridge 52, seed set from the offline query-independent planner). The table reports one readable RBF trade-off point per robot–difficulty stratum: among available budget points within 8% of the shortest audited mean path, it first minimizes online time per query and then the ten-query amortized time; the full-success condition is enforced by selection and recorded in the manifest. Figure 5 shows the complete RBF box-budget curve. PRM is built once per saved obstacle scene and queried on the same ten pairs; RRTConnect and BIT\* solve each query independently as one-shot online baselines. For PRM and BIT\*, the table does not charge one fixed checkpoint to every query. Instead, for each saved

TABLE II

Shelf+IIWA RBF ablation at the selected b100 point. Numeric entries report only the [Q1,Q3] interval and exclude the fixed 0.01s final simplification from timing columns.  $L/L^*$  is the success-only path-length ratio normalized by the best median path for the same shelf query. Seg. is the raw segment fraction.

| Case                     | Build         | Online/q      | Amort@5       | $L/L^*$     | Seg.        |
|--------------------------|---------------|---------------|---------------|-------------|-------------|
| RBF-SH d23               | [0.073,0.077] | [0.030,0.034] | [0.045,0.050] | [0.98,1.07] | [0.14,0.24] |
| CritSample endpoints     | [0.039,0.041] | [0.033,0.035] | [0.041,0.043] | [0.98,1.04] | [0.10,0.18] |
| Link AABB                | [0.056,0.059] | [0.031,0.037] | [0.042,0.048] | [0.99,1.14] | [0.17,0.33] |
| No external LECT, HIFK-5 | [6.771,6.959] | [0.065,0.072] | [1.419,1.461] | [0.98,1.07] | [0.11,0.20] |
| SH w/o broadphase        | [0.124,0.136] | [0.044,0.052] | [0.069,0.079] | [0.98,1.07] | [0.14,0.24] |

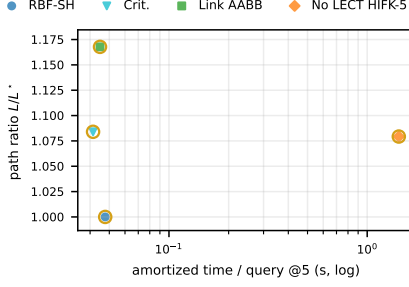


Fig. 3. Shelf+IIWA RBF ablation. Five-query amortized time versus path ratio  $L/L^*$ ; only full-success points are shown; gold rings mark Table II rows.

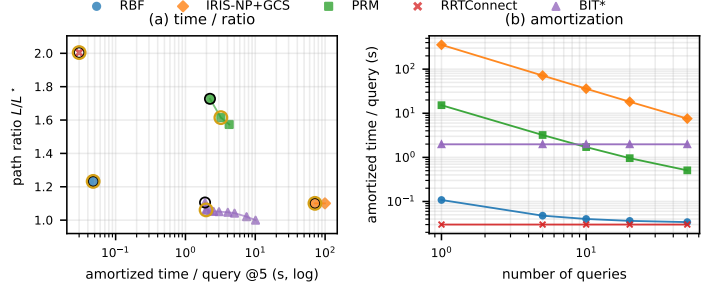


Fig. 4. Shelf+IIWA cross-algorithm trade-off. Panel (a) shows full-success amortized time-path-ratio points; black rings mark first full-success points; panel (b) shows Table III amortization.

TABLE III

Shelf+IIWA per-query cross-algorithm comparison. Rows are stratified by query. Numeric entries report only the [Q1,Q3] interval; online solve time is in seconds and excludes the fixed 0.01s final simplification.  $L/L^*$  is the success-only path-length ratio normalized by the best median path for the same query. RBF uses the registered Exp. 4 profile, with time measured by independent single-query wall-clock reruns.

PRM grows one cumulative shared roadmap for the five-query batch; registered endpoint milestones and cumulative construction are charged to Build. BIT\* reports the best audited incumbent along a single cumulative checkpoint trace. Full curves appear in Figure 4.

| Query | RBF b100<br>Build 0.076s |             | IRIS-NP+GCS<br>Build 355.425s |             | PRM<br>Build 15.032s |             | RRTConnect<br>Build 0.000s |             | BIT*<br>Build 0.000s |             |
|-------|--------------------------|-------------|-------------------------------|-------------|----------------------|-------------|----------------------------|-------------|----------------------|-------------|
|       | $T$ (s)                  | $L/L^*$     | $T$ (s)                       | $L/L^*$     | $T$ (s)              | $L/L^*$     | $T$ (s)                    | $L/L^*$     | $T$ (s)              | $L/L^*$     |
| AS→TS | [0.048,0.058]            | [1.01,1.27] | [0.449,0.449]                 | [1.08,1.08] | [0.029,0.170]        | [1.08,1.31] | [0.024,0.052]              | [1.54,2.05] | [0.204,0.204]        | [0.96,1.08] |
| TS→CS | [0.026,0.030]            | [0.99,1.01] | [0.269,0.269]                 | [1.11,1.11] | [0.123,0.394]        | [2.13,3.10] | [0.021,0.089]              | [2.48,4.51] | [1.904,1.904]        | [1.05,1.45] |
| CS→LB | [0.061,0.065]            | [1.24,1.50] | [0.381,0.381]                 | [1.00,1.00] | [0.184,0.338]        | [1.30,1.95] | [0.027,0.070]              | [1.64,2.35] | [0.054,0.064]        | [1.18,1.72] |
| LB→RB | [0.022,0.027]            | [1.00,1.14] | [0.786,0.786]                 | [1.00,1.00] | [0.100,0.288]        | [1.03,1.20] | [0.001,0.002]              | [1.18,1.23] | [0.054,0.054]        | [0.96,1.02] |
| RB→AS | [0.013,0.015]            | [1.02,1.03] | [0.527,0.527]                 | [1.00,1.00] | [0.253,0.400]        | [1.15,1.22] | [0.001,0.001]              | [1.02,1.08] | [0.054,0.055]        | [1.01,1.02] |

TABLE IV

Saved-catalog random-scene best trade-off points. Numeric entries report only the [Q1,Q3] interval; online solve time is in seconds and excludes the fixed 0.01s final simplification.  $L/L^*$  is the success-only path-length ratio normalized by the best median path for the same saved query when query-level samples are available; summary-only legacy rows use the scenario-level reference. RBF and PRM are reusable planners and report Build/Online/q/ $L/L^*$ ; RRTConnect and BIT\* are one-shot online baselines. PRM and BIT\* entries select the fastest audited checkpoint within  $1.08\times$  of the best path independently for each query/seed, rather than charging one fixed checkpoint to all queries. Full curves appear in Figure 5.

| Scenario     | RBF           |               |             | PRM             |               |             | RRTConnect    |             | BIT*          |             |
|--------------|---------------|---------------|-------------|-----------------|---------------|-------------|---------------|-------------|---------------|-------------|
|              | Build         | Online/q      | $L/L^*$     | Build           | Online/q      | $L/L^*$     | Online/q      | $L/L^*$     | Online/q      | $L/L^*$     |
| IIWA-Medium  | [0.921,1.086] | [0.052,0.119] | [1.00,1.00] | [0.326,1.152]   | [0.012,0.035] | [1.05,1.19] | [0.002,0.006] | [1.01,1.10] | [0.004,0.107] | [1.22,1.61] |
| IIWA-Hard    | [1.076,1.288] | [0.116,0.245] | [1.00,1.02] | [60.007,60.105] | [0.040,0.122] | [1.00,1.08] | [0.004,0.018] | [1.01,1.10] | [0.026,0.774] | [1.27,1.62] |
| UR5-Medium   | [0.284,0.316] | [0.026,0.058] | [1.02,1.15] | [0.317,1.228]   | [0.053,0.061] | [1.02,1.09] | [0.003,0.012] | [1.06,1.41] | [0.007,0.022] | [1.00,1.01] |
| UR5-Hard     | [0.276,0.313] | [0.056,0.079] | [1.02,1.19] | [0.108,0.569]   | [0.052,0.060] | [1.02,1.13] | [0.004,0.021] | [1.15,1.44] | [0.012,0.048] | [1.00,1.02] |
| PANDA-Medium | [0.386,0.669] | [0.030,0.054] | [1.00,1.05] | [0.112,0.839]   | [0.019,0.053] | [1.02,1.08] | [0.002,0.008] | [1.07,1.31] | [0.007,0.032] | [1.00,1.02] |
| PANDA-Hard   | [0.646,0.722] | [0.053,0.111] | [1.00,1.10] | [0.108,0.442]   | [0.011,0.024] | [1.02,1.11] | [0.005,0.042] | [1.06,1.30] | [0.012,0.070] | [1.00,1.05] |

query/seed it selects the fastest strict-audited checkpoint whose path is within 8% of that query's best audited path, then aggregates those selected per-query records. When the short checkpoint traces do not contain a full-success point for a stratum, we add one supplemental 60 s timeout point only for that stratum; the table still reports the

observed return time, not the timeout cap. IRIS-NP+GCS is not included in the random-scene table because its available saved-catalog runs did not achieve sufficiently reliable strict-audit success across the q10x10 random scenes; we therefore keep it only in the Shelf+IIWA contextual comparison where the legacy protocol is directly

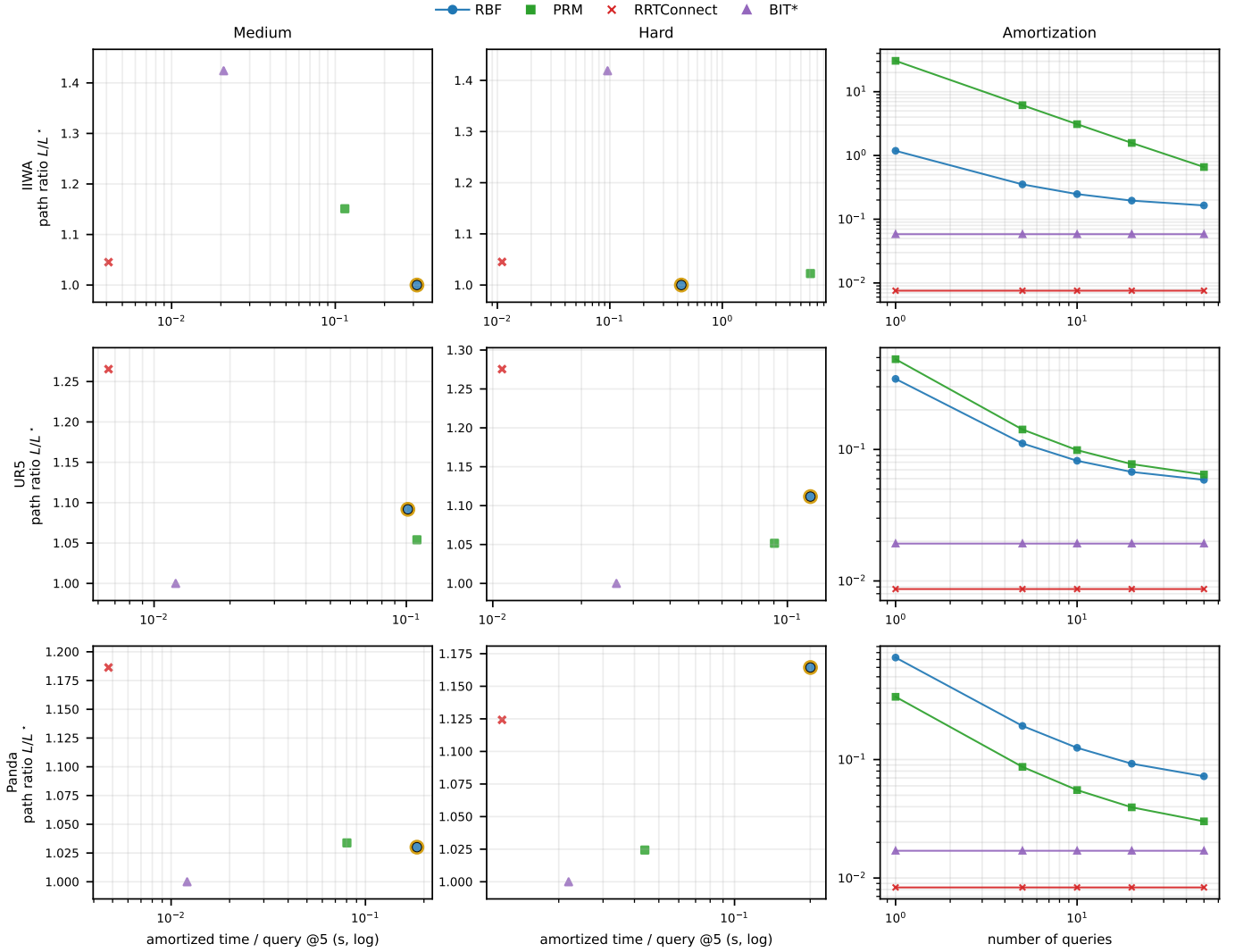


Fig. 5. Saved-catalog random-scene trade-off. Rows separate IIWA, UR5, and Panda; columns show medium, hard, and per-robot query-count amortization. The medium/hard panels plot five-query amortized time versus  $L/L^*$ . Blue curves are RBF budget sweeps on the saved  $q_{10 \times 10}$  prefix catalog. PRM and BIT\* are saved-catalog cumulative reruns with the same 0.01 rad final audit; their reported context points are selected independently per query/seed from the cumulative checkpoints. Only 100% success-rate points are plotted; black rings mark the first full-success point on each curve and gold rings mark the tabulated trade-off point. Context rows for non-RBF reusable planners are shown when available under the same saved-scene protocol.

applicable. These results should be read as a reusable-planner study: RBF’s advantage is expected to appear in online reuse and amortized multi-query cost, while BIT\* and RRTConnect can still be stronger one-shot planners on individual random queries.

### D. Mechanism Studies

TABLE V

Endpoint envelope source study at representative interval widths. Vol. is median workspace envelope volume; time is endpoint-envelope construction only. Negative gap is measured against the sampling-union reference.

| Width | Source     | Safe | Vol.    | Time ( $\mu$ s) | Neg. gap |
|-------|------------|------|---------|-----------------|----------|
| 0.02  | IFK_AA     | Y    | 4.45e-5 | 5.48            | 0        |
| 0.02  | HIFK_3     | Y    | 4.25e-5 | 25.94           | 0        |
| 0.02  | HIFK_5     | Y    | 4.19e-5 | 70.40           | 0        |
| 0.02  | CritSample | N    | 4.09e-5 | 8.68            | -4.8e-5  |
| 0.02  | Analytical | Y    | 4.09e-5 | 6429.96         | 0        |
| 0.02  | MC         | N    | 2.61e-5 | 2251.74         | -6.1e-3  |
| 0.10  | IFK_AA     | Y    | 7.71e-3 | 6.45            | 0        |
| 0.10  | HIFK_3     | Y    | 6.19e-3 | 29.16           | 0        |
| 0.10  | HIFK_5     | Y    | 5.75e-3 | 73.82           | 0        |
| 0.10  | CritSample | N    | 5.09e-3 | 10.03           | -1.2e-3  |
| 0.10  | Analytical | Y    | 5.09e-3 | 6602.25         | 0        |
| 0.10  | MC         | N    | 3.74e-3 | 11233.26        | -3.3e-2  |
| 0.50  | IFK_AA     | Y    | 5.60e0  | 7.35            | 0        |
| 0.50  | HIFK_3     | Y    | 2.03e0  | 29.38           | 0        |
| 0.50  | HIFK_5     | Y    | 1.35e0  | 72.32           | 0        |
| 0.50  | CritSample | N    | 6.40e-1 | 16.33           | -3.1e-2  |
| 0.50  | Analytical | Y    | 6.51e-1 | 7096.01         | -4.5e-3  |
| 0.50  | MC         | N    | 5.34e-1 | 56835.57        | -1.2e-1  |

TABLE VI

Link envelope representation study. Env. reports envelope construction time only; Coll. reports obstacle collision time.

TABLE VIII

Saved-catalog dynamic-update results. Update time is compared with a fresh warm rebuild on the target scene; final audit time is excluded.

| Transition   | SR  | Update (s) | Warm (s) | Speedup |
|--------------|-----|------------|----------|---------|
| easy->medium | 8/8 | 0.026      | 0.412    | 36.085  |
| hard->medium | 8/8 | 0.000      | 0.396    | 907.978 |
| medium->easy | 8/8 | 0.000      | 0.211    | 749.683 |
| medium->hard | 8/8 | 0.005      | 0.420    | 102.821 |

| Width | Envelope    | Splits | Vol. Env. ( $\mu$ s) | Coll. ( $\mu$ s) |
|-------|-------------|--------|----------------------|------------------|
| 0.02  | LinkIAABB   | 1      | 0.013                | 0.110            |
| 0.02  | SupportHull | 1      | 0.000                | 0.166            |
| 0.05  | LinkIAABB   | 1      | 0.020                | 0.139            |
| 0.05  | SupportHull | 1      | 0.003                | 0.177            |
| 0.10  | LinkIAABB   | 1      | 0.038                | 0.141            |
| 0.10  | SupportHull | 1      | 0.014                | 0.188            |
| 0.20  | LinkIAABB   | 1      | 0.147                | 0.136            |
| 0.20  | SupportHull | 1      | 0.105                | 0.171            |
| 0.30  | LinkIAABB   | 1      | 0.502                | 0.137            |
| 0.30  | SupportHull | 1      | 0.440                | 0.175            |
| 0.50  | LinkIAABB   | 1      | 4.514                | 0.143            |
| 0.50  | SupportHull | 1      | 4.448                | 0.184            |

TABLE VII

LECT snapshot/cache operation costs. Exact lookup and endpoint lookup are measured on the persisted d23 snapshot.

| Operation             | Ops   | Avg. ( $\mu$ s) | Nodes   | Evidence |
|-----------------------|-------|-----------------|---------|----------|
| Open snapshot         | 1     | 86.219          | 2097151 | 2097151  |
| Node box              | 20000 | 1.084           | 2097151 | 2097151  |
| Exact lookup          | 20000 | 0.849           | 2097151 | 2097151  |
| Endpoint lookup       | 20000 | 1.860           | 2097151 | 2097151  |
| Endpoint lookup (hot) | 20000 | 0.577           | 2097151 | 2097151  |
| Range query           | 20000 | 22.596          | 2097151 | 2097151  |
| Evidence read         | 20000 | 1.262           | 2097151 | 2097151  |
| Evidence read (hot)   | 20000 | 0.132           | 2097151 | 2097151  |

The mechanism experiments isolate the geometric and cache components from the planner. The endpoint-envelope study compares endpoint sources used by the link-envelope layer; its MC row uses a fixed sample density per joint-box geometric-mean width, and its negative-gap column reports the worst shortfall against the CritSample/Analytical/MC sampling-union reference. Analytical is run with the full phase-3 pipeline even at narrow widths, so the table reflects the certified analytical cost rather than a narrow-box shortcut. The link-envelope study fixes split count  $S = 1$  and compares LinkIAABB, SupportHull, and staged cascades. The LECT study reports read-snapshot operation costs on a persisted robot cache, separating kinematics-keyed evidence replay from scene-dependent free-box labels.

### E. Dynamic Updates

The dynamic-update experiment uses saved random scenes and deterministic obstacle insertion/deletion transitions. Incremental updates are compared against a fresh warm rebuild using the same leaf-sweep-RRT planning profile. The reported metrics separate invalidated boxes, promoted domains, update time, warm rebuild time, validated-query success, and raw segment fraction. All 32 edited-scene queries pass strict audit. One medium-to-hard insertion case requires query-bridge recovery after endpoint segment recovery does not produce an audited

path; this recovery is charged to update time. The median raw segment fraction remains zero for every transition.

## VII. Conclusion

This paper presented RBF, a link-envelope-guided box-forest planner for multi-query manipulation. RBF lifts endpoint-to-link swept-capsule bounds from collision detection to  $C$ -space box validation, grows accepted boxes into a scene-level corridor graph, and separates that scene cache from LECT’s lifelong kinematics-keyed envelope evidence. The design trades convex-region expressivity for low-cost validation, simple graph maintenance, and explicit reuse boundaries.

The guarantee is conditional: paths are certified only when they remain inside a union of conservatively validated boxes (theorem 1). Finite-budget forests need not cover all feasible passages, and repaired or post-processed paths require query validation. Future work will study broader scene distributions, stronger cache transfer, containment-preserving smoothing, and deeper comparisons with high-quality convex-region planners.

## References

- [1] R. Deits and R. Tedrake, “Computing large convex regions of obstacle-free space through semidefinite programming,” in *Algorithmic Foundations of Robotics XI*, 2015, pp. 109–124.
- [2] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, “Motion planning around obstacles with convex optimization,” *Sci. Robot.*, vol. 8, no. 84, 2023.
- [3] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM J. Optim.*, vol. 34, no. 1, pp. 507–532, 2024.
- [4] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.
- [5] —, “Rapidly-exploring random trees: A new tool for path planning,” Iowa State University, Tech. Rep. TR 98-11, 1998.
- [6] L. E. Kavradi, P. Švestka, J.-C. Latombe, and M. H. Overmars, “Probabilistic roadmaps for path planning in high-dimensional configuration spaces,” *IEEE Trans. Robot. Autom.*, vol. 12, no. 4, pp. 566–580, 1996.
- [7] S. Karaman and E. Frazzoli, “Sampling-based algorithms for optimal motion planning,” *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, 2011.
- [8] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators,” in *Algorithmic Foundations of Robotics XV*. Springer, 2022, pp. 328–348.
- [9] M. Petersen and R. Tedrake, “Growing convex collision-free regions in configuration space using nonlinear programming,” 2023. [Online]. Available: <https://arxiv.org/abs/2303.14737>
- [10] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 10 359–10 365.
- [11] D. Blackmore and M. C. Leu, “Analysis of swept volume via Lie groups and differential equations,” *Int. J. Robot. Res.*, vol. 11, no. 6, pp. 516–537, 1992.
- [12] K. Abdel-Malek, J. Yang, D. Blackmore, and K. Joy, “Swept volumes: foundation, perspectives, and applications,” *Int. J. Shape Model.*, vol. 12, no. 1, pp. 87–127, 2006.
- [13] S. Gottschalk, M. C. Lin, and D. Manocha, “OBBTree: A hierarchical structure for rapid interference detection,” in *ACM SIGGRAPH*, 1996.
- [14] E. Larsen, S. Gottschalk, M. C. Lin, and D. Manocha, “Fast proximity queries with swept sphere volumes,” University of North Carolina at Chapel Hill, Tech. Rep. TR99-018, 1999.

- [15] E. G. Gilbert, D. W. Johnson, and S. S. Keerthi, "A fast procedure for computing the distance between complex objects in three-dimensional space," *IEEE J. Robot. Autom.*, vol. 4, no. 2, pp. 193–203, 1988.
- [16] G. v. d. Bergen, "A fast and robust GJK implementation for collision detection of convex objects," *J. Graph. Tools*, vol. 4, no. 2, pp. 7–25, 1999.
- [17] J. Pan, S. Chitta, and D. Manocha, "FCL: A general purpose library for collision and proximity queries," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2012, pp. 3859–3866.
- [18] R. E. Moore, R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*. SIAM, 2009.
- [19] L. Jaulin, M. Kieffer, O. Didrit, and E. Walter, *Applied Interval Analysis*. Springer, 2001.
- [20] J. M. Snyder, "Interval analysis for computer graphics," in *Proc. ACM SIGGRAPH*, 1992, pp. 121–130.
- [21] J.-P. Merlet, "Solving the forward kinematics of a Gough-type parallel manipulator with interval analysis," *Int. J. Robot. Res.*, vol. 23, no. 3, pp. 221–235, 2004.
- [22] —, "Interval analysis for certified numerical solution of problems in robotics," *Int. J. Appl. Math. Comput. Sci.*, vol. 19, no. 3, pp. 399–412, 2009.
- [23] S. Redon, Y. J. Kim, M. C. Lin, and D. Manocha, "Interactive and continuous collision detection for avatars in virtual environments," in *IEEE Virtual Reality Conference*, 2004, pp. 117–124.
- [24] S. Redon, M. C. Lin, D. Manocha, and Y. J. Kim, "Fast continuous collision detection for articulated models," *J. Comput. Inf. Sci. Eng.*, vol. 5, no. 2, pp. 126–137, 2005.
- [25] X. Zhang, S. Redon, M. Lee, and Y. J. Kim, "Continuous collision detection for articulated models using Taylor models and temporal culling," *ACM Trans. Graph.*, vol. 26, no. 3, p. 15, 2007.
- [26] F. Schwarzer, M. Saha, and J.-C. Latombe, "Adaptive dynamic collision checking for single and multiple articulated robots in complex environments," *IEEE Trans. Robot.*, vol. 21, no. 3, pp. 338–353, 2005.
- [27] J. A. Corrales, F. A. Candelas, and F. Torres, "Safe human-robot interaction based on dynamic sphere-swept line bounding volumes," *Robot. Comput.-Integr. Manuf.*, vol. 27, no. 1, pp. 177–185, 2011.
- [28] R. Bohlin and L. E. Kavraki, "Path planning using lazy PRM," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 521–528.
- [29] C. M. Dellin and S. S. Srinivasa, "A unifying formalism for shortest path problems with expensive edge evaluations via lazy best-first search over paths with edge selectors," in *International Conference on Automated Planning and Scheduling (ICAPS)*, 2016.
- [30] E. Schmerling, L. Janson, and M. Pavone, "Optimal sampling-based motion planning under differential constraints: The driftless case," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015.
- [31] D. Coleman, I. Sucan, S. Chitta, and N. Correll, "Reducing the barrier to entry of complex robotic software: A MoveIt! case study," *J. Softw. Eng. Robot.*, vol. 5, no. 1, pp. 3–16, 2014.
- [32] T. Lozano-Pérez, "Spatial planning: A configuration space approach," *IEEE Trans. Comput.*, vol. C-32, no. 2, pp. 108–120, 1983.
- [33] R. A. Brooks and T. Lozano-Pérez, "A subdivision algorithm in configuration space for findpath with rotation," *IEEE Trans. Syst. Man Cybern.*, vol. SMC-15, no. 2, pp. 224–233, 1985.
- [34] J. J. Kuffner and S. M. LaValle, "RRT-connect: An efficient approach to single-query path planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 995–1001.
- [35] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, "Batch informed trees (BIT\*): Sampling-based optimal planning via the heuristically guided search of implicit random geometric graphs," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 3067–3074.
- [36] J. D. Gammell, T. D. Barfoot, and S. S. Srinivasa, "Informed sampling for asymptotically optimal path planning," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 966–984, 2018.
- [37] L. Janson, E. Schmerling, A. Clark, and M. Pavone, "Fast marching tree: A fast marching sampling-based method for optimal motion planning in many dimensions," *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 883–921, 2015.
- [38] I. A. Şucan, M. Moll, and L. E. Kavraki, "The open motion planning library," *IEEE Robot. Autom. Mag.*, vol. 19, no. 4, pp. 72–82, 2012.
- [39] M. Moll, I. A. Şucan, and L. E. Kavraki, "Benchmarking motion planning algorithms: An extensible infrastructure for analysis and visualization," *IEEE Robot. Autom. Mag.*, vol. 22, no. 3, pp. 96–102, 2015.
- [40] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, "CHOMP: Gradient optimization techniques for efficient motion planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2009, pp. 489–494.
- [41] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, "STOMP: Stochastic trajectory optimization for motion planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 4569–4574.
- [42] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, "Motion planning with sequential convex optimization and convex collision checking," *Int. J. Robot. Res.*, vol. 33, no. 9, pp. 1251–1270, 2014.
- [43] M. Mukadam, J. Dong, X. Yan, F. Dellaert, and B. Boots, "Continuous-time gaussian process motion planning via probabilistic inference," *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1319–1340, 2018.